

Light and Radial Velocity Variations in a Nonradially Oscillating Star

by

W. Dziembowski

Copernicus Astronomical Center, Warsaw

Received March 24, 1977

ABSTRACT

Formulae for light and radial velocity variations are derived in terms of solution of the linear oscillations equations. A possibility of observational identification of the oscillation mode is discussed. It is shown that nonradial oscillations cannot be distinguished from the radial ones on the basis of the amplitude ratios of radial velocity and light variations. Instead, the use of the Baade-Weselink method is recommended. For stars with known radii, this method permits to determine the spherical harmonic order of the oscillation.

In this note we consider oscillations corresponding to an arbitrary spherical harmonic distortion of a star whose equilibrium structure may be treated as spherically symmetric. We assume that oscillation amplitudes are sufficiently low, so that the linear approximation is adequate. Formulae for radial velocity variations in a special case of quadrupole oscillations have already been obtained by Ledoux (1951) and Clement (1956).

With the assumptions made, the oscillations corresponding to various spherical harmonics are independent and in a r, Θ, Φ coordinate system, where $r = 0$ coincides with the star center, we express the angular dependence of oscillating quantities in the form

$$Y_{l,j}(\Theta, \Phi) = P_{l,j}(\cos \Theta) e^{ij\Phi}, \quad (1)$$

where $P_{l,j}$ denote the Legendre functions and $P_l = P_{l,0}$ denote the Legendre polynomials. The temporal dependence of oscillating quantities is assumed in the form $\exp(i\omega t)$.

If the effects of rotation are completely ignored, there is an infinite set of eigenfrequencies ω associated with each value of l and independent of the value of j . Thus, in general, an individual oscillation mode corresponds to an arbitrary linear combination of $2l+1$ spherical harmonics of order l . The lowest order effect of rotation is lifting the above degeneracy. In the coordinate system whose polar axis ($\Theta = 0$) coincides with the rotation axis, a separate mode is associated with each j and the corresponding frequencies are distributed at equal distances around the frequency of the $j = 0$ mode.

Let us consider an oscillation mode which in the coordinate system is associated with spherical harmonic $Y_{l,j}$. It is always possible to specify the phase count in such a way that the perturbation of stellar radius, R , is given by

$$\Delta R = R\varepsilon P_{l,j}(\cos\Theta)\cos(\omega t + j\Phi), \quad (2)$$

where ε remains arbitrary except that for the validity of linearization we require $\varepsilon \ll 1$.

The r -component of the surface velocity field is given by $V_r = \partial\Delta R/\partial t$, while the horizontal components have to be found, in principle, from the numerical solution. However, if the gravitational potential perturbation is neglected, which is strictly valid for $l = 1$ and approximately valid for $l > 1$, then the outer boundary condition (see. *e.g.* Dziembowski 1977) leads to

$$V_\Theta = \alpha_H \frac{\partial V_r}{\partial \Theta}, \quad V_\Phi = \frac{\alpha_H}{\sin\Theta} \frac{\partial V_r}{\partial \Phi}, \quad (3)$$

where $\alpha_H = GM/\omega^2 R^3$, G is the gravitational constant and M is the stellar mass. For frequencies corresponding to fundamental radial mode in the case of stars lying on the main sequence or close to it $\alpha_H \approx 0.08$. The same is approximately valid for the fundamental nonradial modes corresponding to low l . For fundamental modes and $l \rightarrow \infty$, we have $\alpha_H \rightarrow 1/l$.

The variable radial velocity, V_{rad} , will be identified here with the surface velocity component parallel to the line of sight, averaged over the visible stellar disc according to the limb darkening law.

It should be noted that because of the line asymmetry, the radial velocity measured by observers does not coincide with the mean radial velocity. This effect in the case of radial pulsations was first discussed by Parsons (1972), who found that the observed velocity is by few percent higher, with the exact figure depending on the star and the method of measurement. Because of considerable difficulties with the extension of Parsons' work into the nonradial oscillation case, we shall limit ourselves

to considering only the mean radial velocity, hoping that also in the present case the difference is not large.

To find variations of the bolometric magnitude, M_{bol} , we have to know variations of the local emergent flux, $\mathcal{F}_n$, which may be written in the following form

$$\Delta \mathcal{F}_n = \mathcal{F}_n \varepsilon f P_{l,j}(\cos \Theta) \cos(\omega t + \psi + j\Phi). \tag{4}$$

We assume that we know f and ψ directly from the solution of non-adiabatic oscillation equations. In addition to the averaged emergent flux variations, the observed luminosity changes are caused by variations of surface elements areas and inclinations. The latter results not only in a change of the projected area of the elements, but also, due to the angular dependence of radiation intensity, in a change of their brightness.

The details of formulae derivation are given in the Appendix. The final result is as follows:

$$V_{rad} = \omega R \varepsilon P_{l,j}(\cos \Theta_0) \cos(\omega t - \pi/2 + j\Phi_0) (u_l + \alpha_H v_l), \tag{5}$$

$$\begin{aligned} \Delta M_{bol} = -1.084 \varepsilon P_{l,j}(\cos \Theta_0) [f \cos(\omega t + \psi + j\Phi_0) b_l + \\ + \cos(\omega t + j\Phi_0) (2b_l - c_l)], \end{aligned} \tag{6}$$

where Θ_0 , Φ_0 determine the direction to the observer,

$$u_l = \int_0^1 h \mu^2 P_l(\mu) d\mu,$$

$$v_l = l \int_0^1 h (P_{l-1} - \mu P_l) \mu^2 d\mu,$$

$$b_l = \int_0^1 h \mu P_l d\mu,$$

$$c_l = l \int_0^1 \left(h + \mu \frac{dh}{d\mu} \right) (P_{l-1} - \mu P_l) d\mu,$$

$h = h(\mu)$ denotes the limb-darkening law normalized in such a way that $\int_0^1 h \mu d\mu = 1$.

It is important to notice that the $V_{rad}/\Delta M_{bol}$ ratio is independent of aspect and that this remains true if, in the coordinate system, oscillations correspond to an arbitrary combination of spherical harmonics of order l . Consequently, a straightforward generalization of the Baade-Weselink formula for stellar radius is possible. The phase of zero effective temperature variation is in our notation $\omega t + j\Phi_0 + \psi = (n + \frac{1}{2})\pi$. If we denote by D_1 and

$(\Delta M_{bol})_1$ the values of radial velocity integral and bolometric magnitude variation, respectively, at this phase we have from Eqs. (5) and (6)

$$R = d_l \frac{D_1}{(\Delta M_{bol})_1}, \quad d_l = \frac{1.084(2b_l - c_l)}{u_l + \alpha_H v_l}. \quad (7)$$

The l -dependent integrals can be evaluated explicitly if h is given in the form

$$h = \sum \chi_m \mu^m$$

in which case they are reduce to the sum of integrals

$$I_{l,m} = \int_0^1 P_l \mu^m d\mu.$$

Explicit formulae for $I_{l,m}$ are given by Erdelyi (1953) for arbitrary m . For m integer the result can be written as follows.

If $l > m$

$$I_{l,m} = \begin{cases} 0 & \text{for } l-m \text{ even,} \\ \frac{(-1)^{\frac{l-m-1}{2}} m! (l-m-1)!}{2^l \left(\frac{l-m-1}{2}\right)! \left(\frac{l+m+1}{2}\right)!} & \text{for } l-m \text{ odd.} \end{cases}$$

If $l \leq m$

$$I_{l,m} = \begin{cases} \frac{2^l m! \left(\frac{m+l}{2}\right)!}{(m+l+1)! \left(\frac{m-l}{2}\right)!} & \text{for } m-l \text{ even} \\ \frac{m! \left(\frac{m-l-1}{2}\right)!}{2^{l+1} (m-l)! \left(\frac{m+l+1}{2}\right)!} & \text{for } m-l \text{ odd.} \end{cases}$$

For $l \rightarrow \infty$ and m finite, we have

$$I_{l,m} = (-1)^{\frac{l-m-1}{2}} m! l^{-(m+3/2)} \sqrt{\frac{2}{\pi}}.$$

In Table 1, the values of l -dependent coefficients are given in the case of the Eddington limb-darkening law $h = 1 + 1.5\mu$. Asymptotic expressions for $l \rightarrow \infty$ in this case are:

Table 1

Coefficients in Eqs. (5) and (6) for the case of the Eddington limb-darkening law

l	u_l	v_l	b_l	c_l
0	7.08E-1	0,0	1.0	0
1	5.50E-1	4.50E-1	7.08E-1	1.416
2	3.21E-1	7.75E-1	3.25E-1	1.95
3	1.27E-1	5.94E-1	6.26E-2	7.50E-1
4	2.34E-2	1.56E-1	-2.08E-2	-4.17E-1
5	-5.21E-3	-1.17E-2	-7.82E-3	-2.34E-1
6	-2.34E-3	-3.19E-2	7.81E-3	3.28E-1
7	1.56E-3	4.37E-2	2.34E-3	1.21E-1
8	5.87E-4	1.41E-2	-3.91E-3	-2.81E-1

or l even

$$u_l = -\frac{9c_l}{l^4}, \quad v_l = -\frac{3c_l}{l^2},$$
$$b_l = \frac{c_l}{l^2}, \quad c_l = (-1)^{(l/2+1)} \sqrt{\frac{2}{\pi l}};$$

for l odd

$$u_l = \frac{2c_l}{3l^2}, \quad v_l = \frac{c_l}{3},$$
$$b_l = \frac{c_l}{l^2}, \quad c_l = (-1)^{(l+1)/2} \frac{1}{l} \sqrt{\frac{2}{\pi l}}.$$

One may see from the above formulae, as well as from the numbers given in Table 1, that these coefficients are small for large l . In particular, one finds that even for the most favorable aspect conditions the observed radial velocity amplitude becomes less than one tenth of the maximum velocity amplitude on stellar surface for $l \geq 4$. Thus, the line broadening effects become dominant already at relatively low-order spherical harmonics.

As an application of the formulae derived, we shall consider a possibility of determining the spherical harmonic order on the basis of light and radial velocity curves for a variable star. In a recent paper, Breger *et al.* (1976) compiled photometric and spectroscopic data for several δ Scuti variables. The data we shall be concerned with are ψ_s -phase differences between the inverse radial velocity and light curve as well as η -ratio of $\Delta R/R$ amplitude and ΔM_{bol} amplitude. Values of η were derived from the radial velocity curve upon using Parsons' (1972) converting factor

for radial pulsation, 1.34, rather than a standard value of 1.41 which is equal $1/u_0$ in our notation.

Values of η for these objects are in the range 0.07-0.14 with average 0.11, while the values of ψ_s are in the range 0.00-0.12 with average 0.09. The authors suggest that the determined values "put considerable constraints on the nonradial modes that are allowed for δ Scuti variables."

A linear nonadiabatic analysis of oscillation properties of the relevant stellar models (Dziembowski 1975, 1977) shows that many radial and non-radial modes may be excited in these variables. For low-order spherical harmonics ($l \lesssim 8$) in the frequency range corresponding to the fundamental mode and first overtones of radial pulsations, f depends mainly on ω and only slightly on l . Spectrum of eigenfrequencies ω for such modes depends on the star internal structure and is considerably denser for $l \geq 1$ than for $l = 0$, provided that the star has a hydrogen exhausted core.

In Table 2, the theoretical η and ψ_s values are given for $l = 0, 1$ and 2 and frequencies equal to those of the fundamental mode and first two

Table 2

Oscillation properties of δ Scuti star models with mass $2M_\odot$ and standard population I composition ($X = 0.7$, $Z = 0.02$)

Mean Model	Period (days)	$l = 0$		$l = 1$		$l = 2$	
		η	ψ_s	η	ψ_s	η	ψ_s
$L = 25L_\odot$ $\log T_e = 3.876$	0.1204	0.058	0.059	0.065	0.076	0.084	0.107
	0.0917	0.053	0.050	0.058	0.066	0.071	0.095
	0.0739	0.052	0.059	0.056	0.076	0.067	0.103
$L = 25L_\odot$ $\log T_e = 3.851$	0.1432	0.081	0.007	0.093	0.034	0.118	0.082
	0.1091	0.081	0.011	0.090	0.038	0.109	0.085
	0.0877	0.087	0.050	0.091	0.076	0.098	0.114
$L = 25L_\odot$ $\log T_e = 3.826$	0.1705	0.116	-0.043	0.141	0.000	0.180	0.076
	0.1300	0.126	-0.015	0.142	0.029	0.159	0.101
	0.1043	0.115	0.049	0.118	0.082	0.123	0.131
$L = 15L_\odot$ $\log T_e = 3.851$	0.0985	0.094	-0.007	0.110	0.026	0.138	0.083
	0.0758	0.097	0.007	0.108	0.039	0.130	0.085
	0.0617	0.095	0.083	0.099	0.080	0.107	0.124

overtones of radial pulsations. For $l > 0$, values of f for $l = 0$ were taken after checking that this procedure introduces a negligible error. The envelope models chosen for these calculations well represent the typical data for δ Scuti variables.

The theoretical values of η agree well with the observed ones for almost all modes included. There is a dependence of η on both l and period for a given model, but it is a weak dependence in comparison with the de-

dependence on the effective temperature. Moreover, on the basis of other calculations, strong dependence of η on the hydrogen content was found. Thus, on the basis of the determination we cannot distinguish the nonradial oscillations corresponding to low-order spherical harmonics from radial pulsations.

A comparison of the theoretical and observed ψ_s values seems more significant. Surprisingly, the present comparison indicates the $l = 2$ rather than $l = 0$ modes excitation. It should be pointed out, however, that the theoretical φ_s values are not quite reliable, because of the dependence of the $\mathcal{F}_n$ phase on the convection treatment (Castor 1971). The present results are obtained with the complete neglect of convection.

A modified Baade-Weselink method appears the only potentially powerful method of the oscillation mode identification based on simultaneous light and radial velocity curves. If the stellar radius is known, Eq. (7) may be used to determine d_l observationally. On the other hand, d_l may be calculated for each hypothesis about the oscillation mode. For the Eddington limb-darkening law and $a_H = 0.08$, which corresponds to the fundamental radial modes for δ Scuti star models, we get: $d_0 = 3.06$, $d_1 = 0$, $d_2 = -3.68$, $d_3 = -3.88$, $d_4 = 11.3$, $d_5 = -38.4$, $d_6 = 69.2$. Except for a possible ambiguity in the case of $l = 2$ and 3, the dependence of d_l on l is very strong. This is why the encouragement is given to applying the Baade-Wesselink method, in spite of the well-known problems in its practical use.

An independent method of mode identification is a comparison of the observed line profile variations with those calculated, upon assuming various spherical harmonic dependences for oscillations. Such a method has already been used for interpretation of the variability (Clement 1956, Christy 1966, Osaki 1971), but no general criteria for the mode identification were developed.

The author would like to thank Professor B. Paczyński for helpful discussions.

REFERENCES

- | | | |
|--|------|--|
| Breger, M., Hutchins, J., and Kuhi,
L. Y. | 1976 | <i>Ap.J.</i> 210 , 163 |
| Castor, J. I. | 1971 | <i>Ap.J.</i> 166 , 109. |
| Christy, R. | 1966 | <i>Astr.J.</i> 72 , 293. |
| Clement, M. J. | 1965 | <i>Ap.J.</i> 141 , 1443. |
| Dziembowski, W. | 1975 | <i>Mem.Soc.Roy.Liège</i> , 6 ^e s, VIII, 287. |
| | 1977 | <i>Acta Astr.</i> 27 , 95. |
| Erdelyi, A. (ed) | 1953 | <i>Higher Transcendental Functions</i> , Bateman Manuscript Project, Vol. I (New York: Mc. Graw-Hill). |

- Ledoux, P. 1951 *Ap.J.* **114**, 373.
 Osaki, Y. 1971 *Publ. Astr. Soc. Japan*, **23**, 485.
 Parsons, S. B. 1972 *Ap.J.* **174**, 57.

APPENDIX

We have assumed that the star performs oscillations corresponding to $Y_{l,j}$ spherical harmonic in the r, Θ, Φ coordinate system. In order to calculate the averaged radial velocity and luminosity variations, it is convenient to consider oscillations in the r, ϑ, φ coordinate system, in which the polar axis coincides with the direction to the observer.

Because of the completeness of the set of $2l+1$ spherical harmonics of order l , we have

$$Y_{l,j}(\Theta, \Phi) = \sum_{k=-l}^l q_k Y_{l,k}(\vartheta, \varphi),$$

where complex coefficients q_k depend only on Θ_0, Φ_0 — the angular coordinates of the observer in the r, Θ, Φ system. Only coefficient q_0 will be needed in an explicit form, which may be readily found upon using the well-known formula:

$$\int_0^{\pi} \int_0^{2\pi} Y_{l,j}(\Theta, \Phi) P_l(\cos \vartheta) d\Phi \sin \Theta d\Theta = Y_{l,j}(\Theta_0, \Phi_0) 2\pi \int_{-1}^1 P_l^2(\mu) d\mu,$$

which leads to

$$q_0 = Y_{l,j}(\Theta_0, \Phi_0). \quad (\text{A1})$$

In the r, ϑ, φ system, the perturbation of stellar radius has the form

$$\Delta R = R\varepsilon \operatorname{Re}(e^{i\omega t} \sum_k q_k Y_{l,k}),$$

and the components of the surface velocity field are

$$V_r = \frac{\partial \Delta R}{\partial t}, \quad V_\vartheta = a_H \frac{\partial V_r}{\partial \vartheta}, \quad V_\varphi = \frac{a_H}{\sin \vartheta} \frac{\partial V_r}{\partial \varphi}.$$

The unit vector directed toward the observer is $(\cos \vartheta, -\sin \vartheta, 0)$ so that the radial velocity of a surface element located at the coordinates ϑ, φ is given by

$$v_{rad} = R\varepsilon \omega \operatorname{Re}[e^{i(\omega t - \pi/2)} \sum_k q_k (Y_{l,k} \cos \vartheta - \frac{\partial Y_{l,k}}{\partial \vartheta} \sin \vartheta)].$$

Neglecting terms of the higher order in ε , we have for a mean radial velocity of the stellar disc

$$V_{rad} = \frac{1}{2\pi} \int_0^1 \left(\int_0^{2\pi} v_{rad} d\varphi \right) h(\mu) \mu d\mu,$$

where $\mu = \cos \vartheta$ and $h(\mu)$ denotes the normalized limb-darkening law.

The final expression for V_{rad} given by Eq. (5) is obtained after using Eq. (A1) and the following relation

$$(1 - \mu^2) \frac{dP_l}{d\mu} = l(P_{l-1} - \mu P_l). \quad (A2)$$

For the total luminosity variation we have

$$\Delta L = \frac{1}{2\pi} \iint [(\Delta \mathcal{F}_n h + \mathcal{F}_n \Delta h) R^2 \mu \sin \vartheta + \Delta S] d\varphi d\vartheta,$$

$\Delta \mathcal{F}_n$ is now given by the following formula

$$\Delta \mathcal{F}_n = \mathcal{F}_n \varepsilon \operatorname{Re}(e^{i(\omega t + \varphi)} \sum_k q_k Y_{l,k}),$$

Δh stands for the change of the brightness of a surface element due to the change of the normal, $\vec{n}$, to the stellar surface; thus, we have

$$\Delta h = - \frac{dh}{d\mu} \Delta n_\vartheta \sin \vartheta,$$

where

$$\Delta n_\vartheta = - \frac{1}{R} \frac{\partial \Delta R}{\partial \vartheta}.$$

$\Delta S d\vartheta d\varphi$ gives the change of the area of a surface element on the plane perpendicular to the line of sight. One easily finds that

$$\Delta S = R \sin \vartheta (2 \Delta R \cos \vartheta + \frac{\partial \Delta R}{\partial \vartheta} \sin \vartheta).$$

Perturbation of the integral boundary gives contribution of the order of ε^2 , so it is not included here.

Thus, for the relative luminosity variation we have

$$\begin{aligned} \frac{\Delta L}{L} = & \varepsilon \{ f \operatorname{Re}(q_0 e^{i(\omega t + \varphi)}) \int P_l h \mu d\mu + \\ & + \operatorname{Re}(q_0 e^{i\omega t}) \int \left[2P_l h \mu + \frac{dP_l}{d\vartheta} \sin \vartheta \left(h + \mu \frac{dh}{d\mu} \right) \right] d\mu \}. \end{aligned}$$

Using again Eqs. (A1) and (A2) one obtains Eq. (6) for the bolometric magnitude variation.